

If Z is not of full rank, then there are infinitely many LSE's of β . It can be shown (exercise) that any LSE of β is of the form

$$\hat{\beta} = (Z^\tau Z)^- Z^\tau X, \quad (3.29)$$

where $(Z^\tau Z)^-$ is called a *generalized inverse* of $Z^\tau Z$ and satisfies

$$Z^\tau Z (Z^\tau Z)^- Z^\tau Z = Z^\tau Z.$$

Generalized inverse matrices are not unique unless Z is of full rank, in which case $(Z^\tau Z)^- = (Z^\tau Z)^{-1}$ and (3.29) reduces to (3.28).

To study properties of LSE's of β , we need some assumptions on the distribution of X . Since Z_i 's are nonrandom, assumptions on the distribution of X can be expressed in terms of assumptions on the distribution of ε . Several commonly adopted assumptions are stated as follows.

Assumption A1: ε is distributed as $N_n(0, \sigma^2 I_n)$ with an unknown $\sigma^2 > 0$.

Assumption A2: $E(\varepsilon) = 0$ and $\text{Var}(\varepsilon) = \sigma^2 I_n$ with an unknown $\sigma^2 > 0$.

Assumption A3: $E(\varepsilon) = 0$ and $\text{Var}(\varepsilon)$ is an unknown matrix.

Assumption A1 is the strongest and implies a parametric model. We may assume a slightly more general assumption that ε has the $N_n(0, \sigma^2 D)$ distribution with unknown σ^2 but a known positive definite matrix D . Let $D^{-1/2}$ be the inverse of the square root matrix of D . Then model (3.25) with assumption A1 holds if we replace X , Z , and ε by the transformed variables $\tilde{X} = D^{-1/2}X$, $\tilde{Z} = D^{-1/2}Z$, and $\tilde{\varepsilon} = D^{-1/2}\varepsilon$, respectively. A similar conclusion can be made for assumption A2.

Under assumption A1, the distribution of X is $N_n(Z\beta, \sigma^2 I_n)$, which is in an exponential family $\mathcal{P}$ with parameter $\theta = (\beta, \sigma^2) \in \mathcal{R}^p \times (0, \infty)$. However, if the matrix Z is not of full rank, then $\mathcal{P}$ is not identifiable (see §2.1.2), since $Z\beta_1 = Z\beta_2$ does not imply $\beta_1 = \beta_2$.

Suppose that the rank of Z is $r \leq p$. Then there is an $n \times r$ submatrix Z_* of Z such that

$$Z = Z_* Q \quad (3.30)$$

and Z_* is of rank r , where Q is a fixed $r \times p$ matrix. Then

$$Z\beta = Z_* Q\beta$$

and $\mathcal{P}$ is identifiable if we consider the reparameterization $\tilde{\beta} = Q\beta$. Note that the new parameter $\tilde{\beta}$ is in a subspace of $\mathcal{R}^p$ with dimension r .

In many applications, we are interested in estimating some linear functions of β , i.e., $\vartheta = l^\tau \beta$ for some $l \in \mathcal{R}^p$. From the previous discussion, however, estimation of $l^\tau \beta$ is meaningless unless $l = Q^\tau c$ for some $c \in \mathcal{R}^r$ so that

$$l^\tau \beta = c^\tau Q\beta = c^\tau \tilde{\beta}.$$

The following result shows that $l^\tau \beta$ is estimable if $l = Q^\tau c$, which is also necessary for $l^\tau \beta$ to be estimable under assumption A1.

Theorem 3.6. Assume model (3.25) with assumption A3.

(i) A necessary and sufficient condition for $l \in \mathcal{R}^p$ being $Q^\tau c$ for some $c \in \mathcal{R}^r$ is $l \in \mathcal{R}(Z) = \mathcal{R}(Z^\tau Z)$, where Q is given by (3.30) and $\mathcal{R}(A)$ is the smallest linear subspace containing all rows of A .

(ii) If $l \in \mathcal{R}(Z)$, then the LSE $l^\tau \hat{\beta}$ is unique and unbiased for $l^\tau \beta$.

(iii) If $l \notin \mathcal{R}(Z)$ and assumption A1 holds, then $l^\tau \beta$ is not estimable.

Proof. (i) Note that $a \in \mathcal{R}(A)$ if and only if $a = A^\tau b$ for some vector b . If $l = Q^\tau c$, then

$$l = Q^\tau c = Q^\tau Z_*^\tau Z_* (Z_*^\tau Z_*)^{-1} c = Z^\tau [Z_* (Z_*^\tau Z_*)^{-1} c].$$

Hence $l \in \mathcal{R}(Z)$. If $l \in \mathcal{R}(Z)$, then $l = Z^\tau \zeta$ for some ζ and

$$l = (Z_* Q)^\tau \zeta = Q^\tau c$$

with $c = Z_*^\tau \zeta$.

(ii) If $l \in \mathcal{R}(Z) = \mathcal{R}(Z^\tau Z)$, then $l = Z^\tau Z \zeta$ for some ζ and by (3.29),

$$\begin{aligned} E(l^\tau \hat{\beta}) &= E[l^\tau (Z^\tau Z)^{-} Z^\tau X] \\ &= \zeta^\tau Z^\tau Z (Z^\tau Z)^{-} Z^\tau Z \beta \\ &= \zeta^\tau Z^\tau Z \beta \\ &= l^\tau \beta. \end{aligned}$$

If $\bar{\beta}$ is any other LSE of β , then, by (3.27),

$$l^\tau \hat{\beta} - l^\tau \bar{\beta} = \zeta^\tau (Z^\tau Z)(\hat{\beta} - \bar{\beta}) = \zeta^\tau (Z^\tau X - Z^\tau X) = 0.$$

(iii) Under assumption A1, if there is an estimator $h(X, Z)$ unbiased for $l^\tau \beta$, then

$$l^\tau \beta = \int_{\mathcal{R}^n} h(x, Z) (2\pi)^{-n/2} \sigma^{-n} \exp \left\{ -\frac{1}{2\sigma^2} \|x - Z\beta\|^2 \right\} dx.$$

Differentiating w.r.t. β and applying Theorem 2.1 lead to

$$l = Z^\tau \int_{\mathcal{R}^n} h(x, Z) (2\pi)^{-n/2} \sigma^{-n-2} (x - Z\beta) \exp \left\{ -\frac{1}{2\sigma^2} \|x - Z\beta\|^2 \right\} dx,$$

which implies $l \in \mathcal{R}(Z)$. ■

Theorem 3.6 shows that LSE's are unbiased for estimable parameters $l^\tau \beta$. If Z is of full rank, then $\mathcal{R}(Z) = \mathcal{R}^p$ and, therefore, $l^\tau \beta$ is estimable for any $l \in \mathcal{R}^p$.